Predicting MLB Pitcher Injury Risk Using Statcast Data

Nana Noda-Morgan

DSC 680 Applied Data Science

Professor Iranitalab

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The Business Problem

PITCHERS ARE THE HIGHEST-PAID PLAYERS IN BASEBALL ($37–42M AVG)

PITCHER INJURIES ARE COSTLY AND DISRUPTIVE (ESPECIALLY UCL TEARS)

Can publicly available pitch data give teams an early warning?

Research Questions

01

Do higher pitch counts mean higher injury risk?

02

Does velocity drop signal imminent IL placement?

03

Does pitch mix diversity matter?

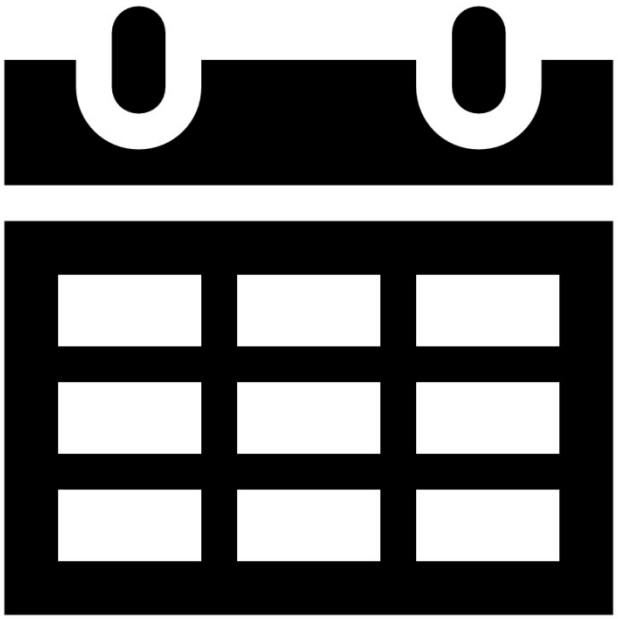
04

Does the role (starter vs. reliever) affect risk?

Data Sources

•Statcast: 3.1 million pitches via pybaseball (2021–2025)
•MLB Stats API: 8,149 pitcher IL transactions (2022–2025)

The original plan was CSV downloads and web scraping.
Both had to be adapted during the project.

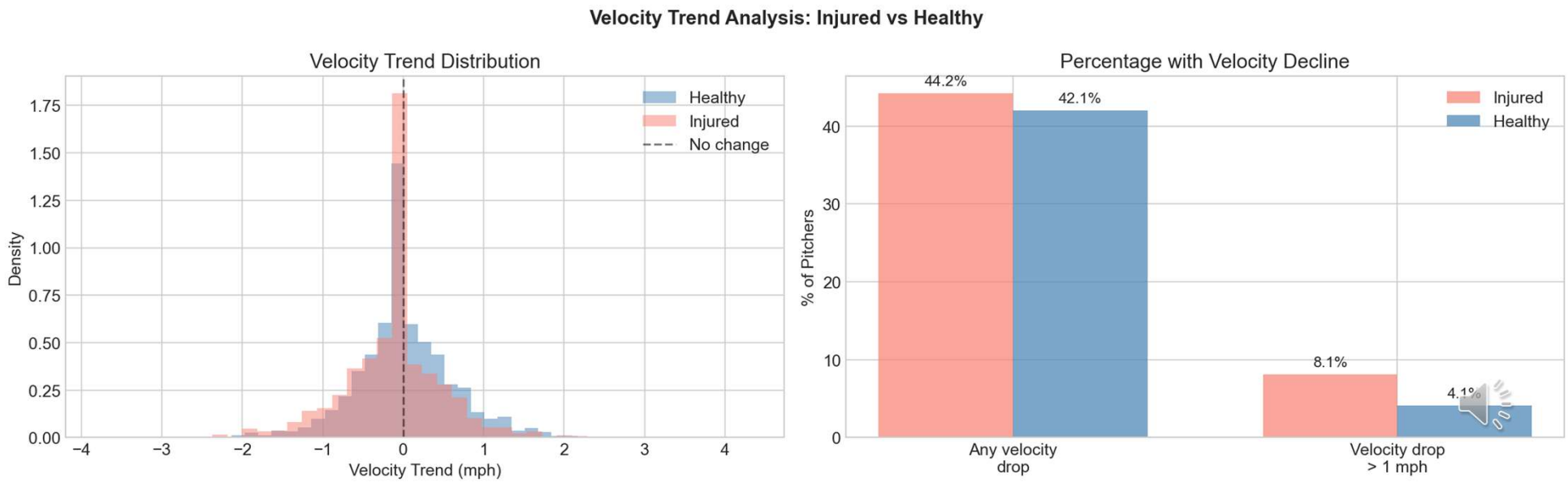


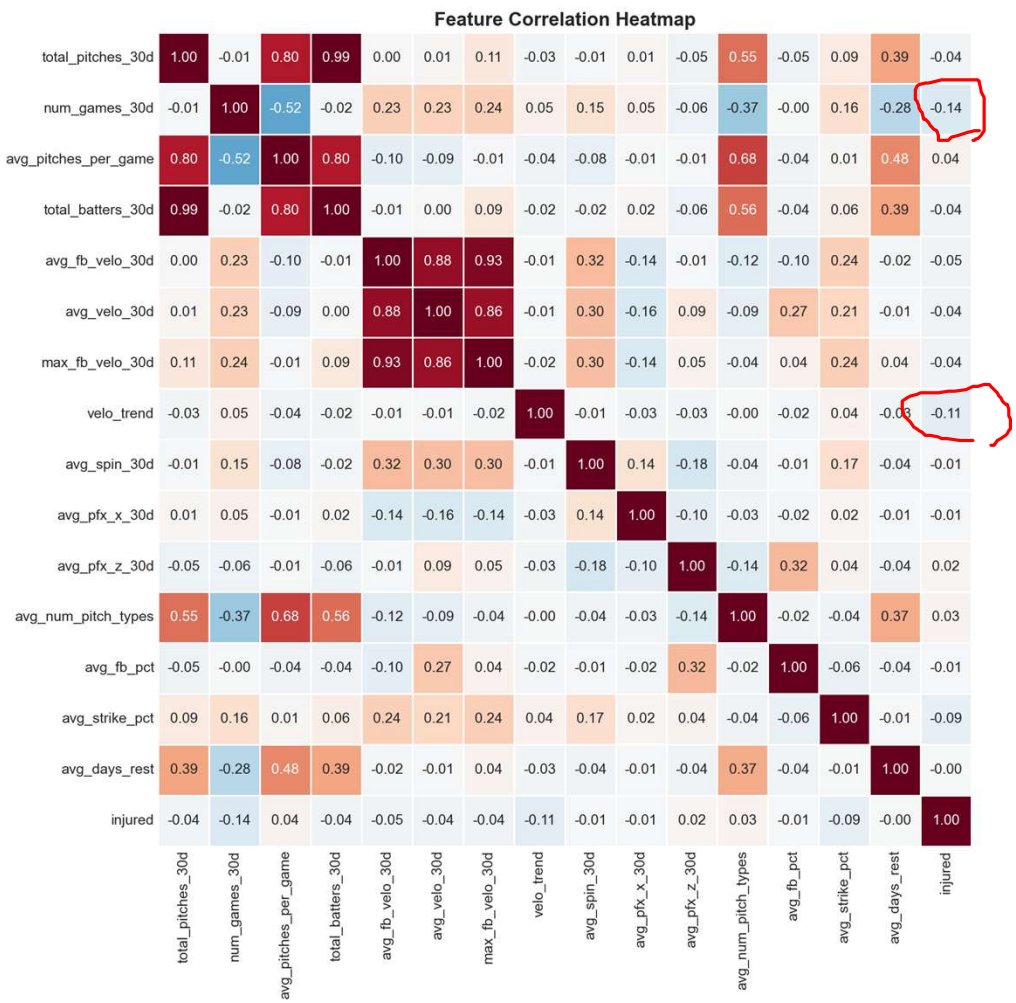
Feature Engineering The 30-day Window

- For each IL placement, look back 30 days and aggregate pitching data
- 16 features: workload, velocity, movement, pitch mix, rest, starter flag
- Control group: healthy windows from both never-injured and eventually-injured pitchers
- Final dataset: 3,730 rows (1,628 injured, 2,102 healthy)

Exploratory Data Analysis

Injured pitchers show declining velocity (-0.11 mph), fewer games, fewer strikes

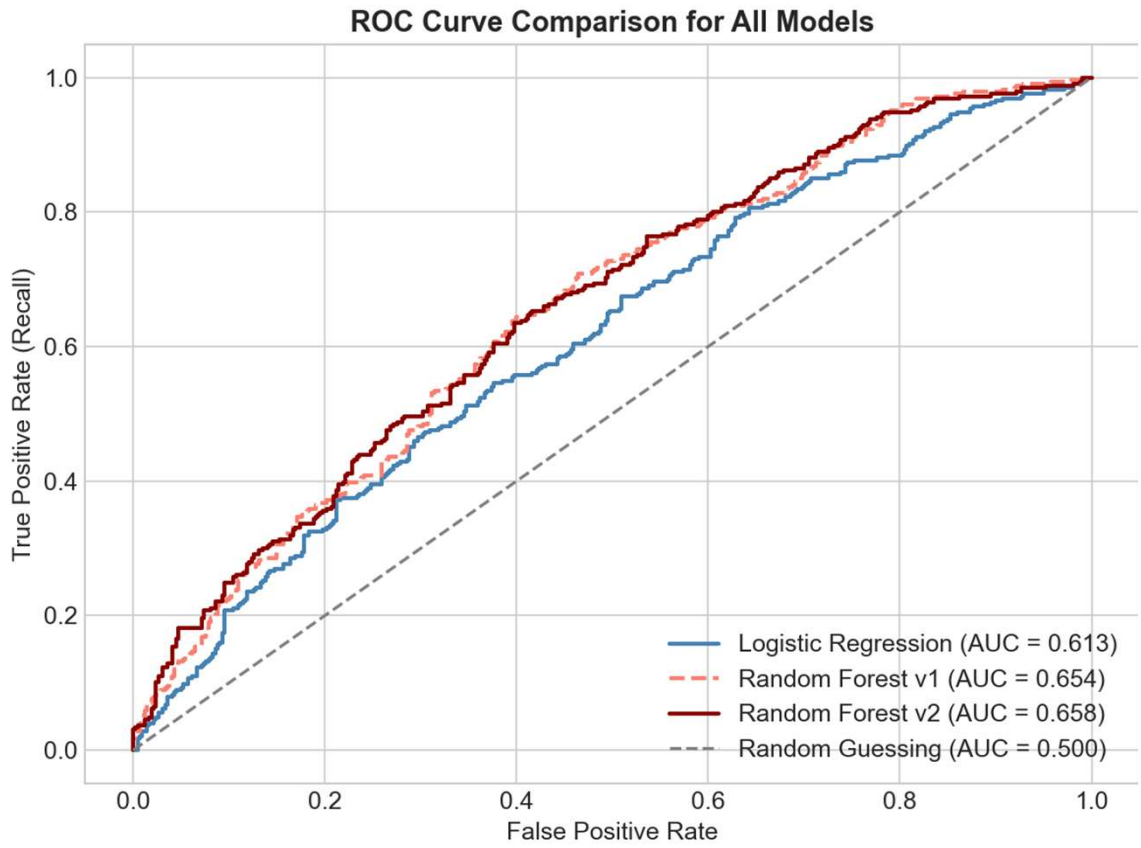




Exploratory Data Analysis (cont'd)

- Strongest correlations with injury: num_games (-0.14), velo_trend (-0.11)

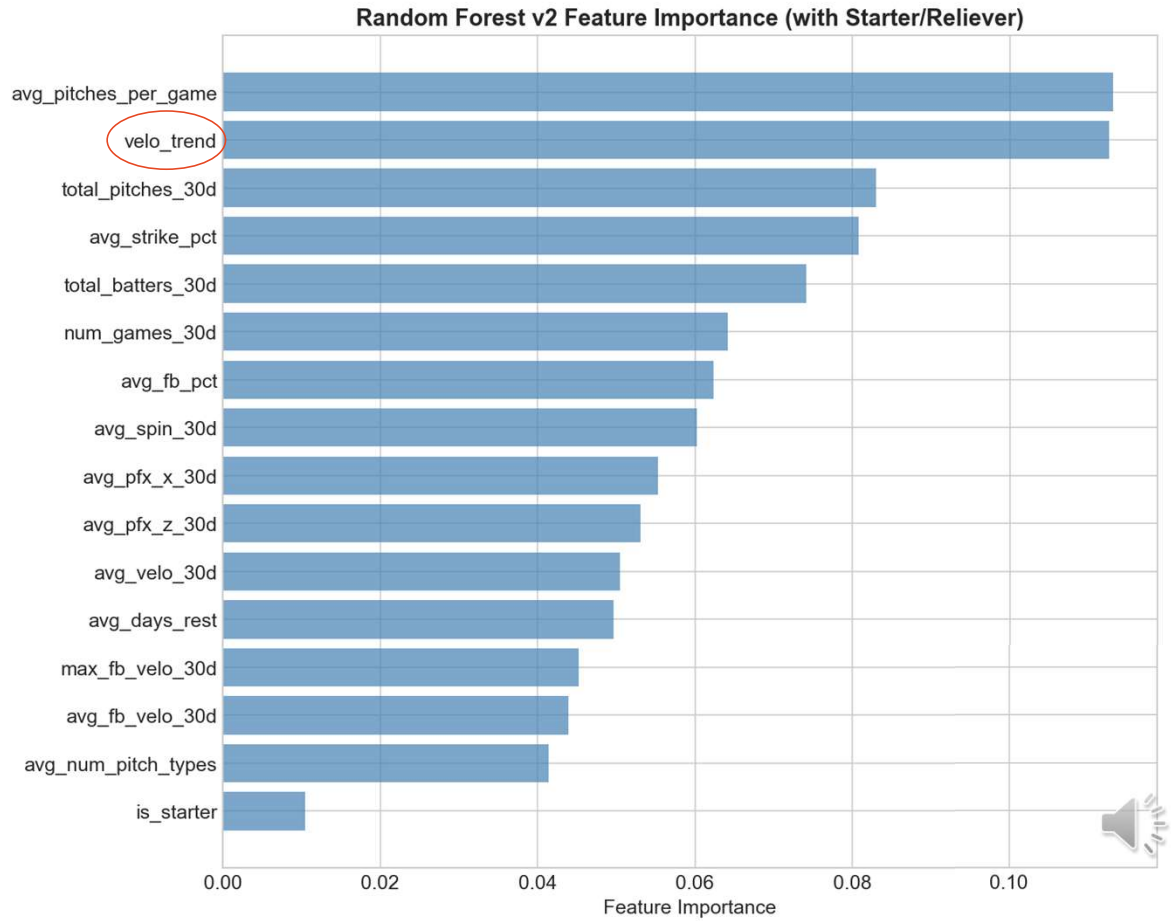
- Most features are weakly correlated indicating a complex prediction problem



Model Results

- Logistic Regression: AUC = 0.61 (baseline)
- Random Forest v1: AUC = 0.65 (non-linear patterns help)
- Random Forest v2: AUC = 0.66 (starter/reliever feature adds value)
- All models stable across cross-validation folds

What Matters Most



Why Only 0.65 AUC?

MANY INJURY FACTORS AREN'T IN STATCAST: TRAINING, BIOMECHANICS, MEDICAL HISTORY, SLEEP, NUTRITION

61% OF IL PLACEMENTS HAD NO PITCHING DATA IN THE 30-DAY WINDOW

AUC OF 0.65 IS MEANINGFUL — NOT A CRYSTAL BALL, BUT BETTER THAN GUESSING

COMPARABLE TO PUBLISHED SPORTS INJURY MODELS

Recommendations

01

Monitor velocity trends. It is the most actionable early warning sign

02

Track per-game workload, not just season totals

03

Use as one input alongside medical evaluation, not standalone

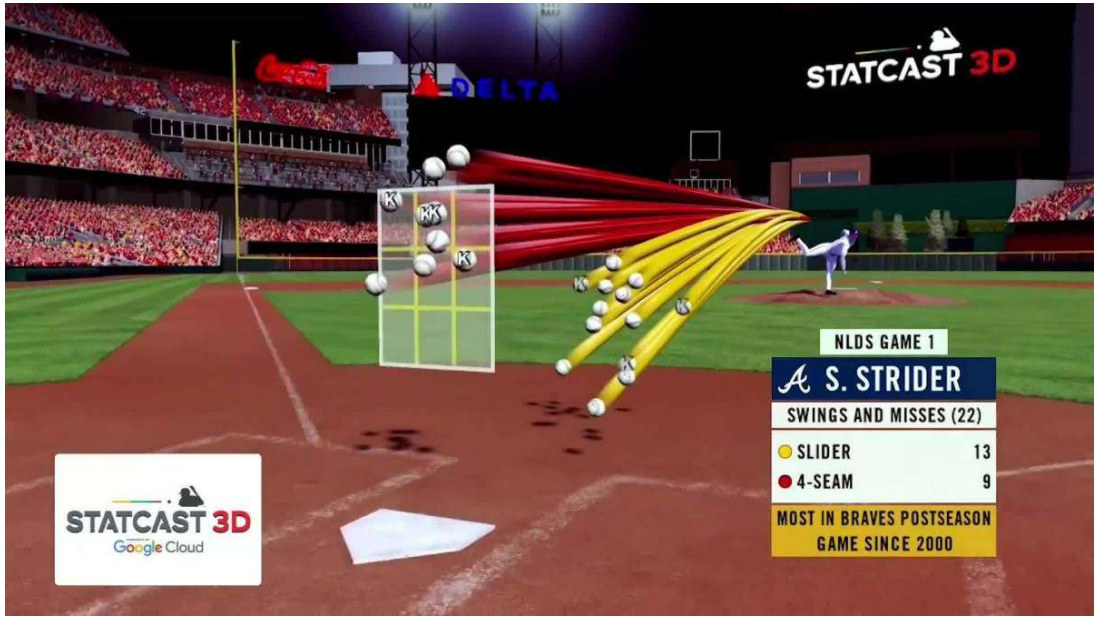
04

Retrain annually with new season data

Implementation Plan

Phase 1	Phase 2	Phase 3
Automate nightly data pipeline during season	Dashboard for coaching/medical staff	Annual retraining and performance tracking

Key Takeaways



- Velocity decline is the clearest pre-injury signal in public data
- Pitch-level data contains real but moderate predictive power
- The model adds value as a supplementary tool, not a replacement for medical judgment
- Framework is reproducible and extensible

Thank you

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Q&A

Q1: Why a 30-day window?

A: The 30 days is a good balance between having enough data points and recency.

Future work should test other durations such as 14, 45, or 60 days.

Q&A

Q2: Why is the AUC only around 0.65? Is that useful?

A: Many injury factors aren’t in Statcast. The results are comparable to published sports injury models. It is useful as a screening tool.

Q&A

Q3: How did you handle the imbalance between injured and healthy pitchers?

A: Two-source control group (healthy pitchers + healthy periods from injured pitchers); Used the class_weight="balanced" parameter in Random Forest.

Q&A

Q4: Could this model be used to predict specific injury types like UCL tears?

A: In principle yes, but smaller sample sizes per injury type make it harder; current approach maximizes statistical power.

Q&A

Q5: How do you classify starters versus relievers?

A: From actual game data: whoever pitched inning 1 was the starter; pitchers starting ≥50% of appearances classified as SP.

Q&A

Q6: Why did you exclude the 2020 season?

A: 60-game COVID season had fundamentally different workload patterns; including it would add noise.

Q&A

Q7: What would improve the model's accuracy the most?

A: Biomechanical data from motion capture or wearable sensors (arm slot, elbow torque, shoulder rotation).

Q&A

Q8: Why were 2,531 IL placements excluded from the analysis?

A: No pitching data in the 30-day window (spring training injuries, repeat IL stints, offseason placements).

Q&A

Q9: Could a team actually use this model during the season?

A: Yes, all features are from real-time Statcast data; model could update after every game.

Q&A

Q10: What is the velocity trend feature and why is it important?

A: Change in fastball velocity from first half to second half of the 30-day window; most statistically significant differentiator ($p < 0.001$); observable in real-time.